# Multimodal Text-to-Set Remote Sensing Image Generation With Anchor Diffusion

Yuhan Kang, Honghu Pan, Hao Liu, *Graduate Student Member, IEEE*, Qiang Liu, Weiying Xie, *Senior Member, IEEE*, Leyuan Fang, *Senior Member, IEEE*, and Lorenzo Bruzzone, *Fellow, IEEE*

***Abstract*—Multimodal remote sensing imagery provides complementary information for Earth observation, yet acquiring large-scale, spatially aligned multimodal data is expensive and labor-intensive, motivating data-efficient multimodal image generation. Existing methods are primarily developed for single-modal synthesis or pairwise cross-modal generation and thus cannot explicitly model the scene structure shared by multiple modalities for multimodal set generation. This limitation makes it difficult to simultaneously preserve cross-modal spatial alignment and modality-specific fidelity. To address these issues, we propose anchor diffusion (AnchorDiff), a unified framework that is able to generate spatially aligned multimodal remote sensing images from textual descriptions. To enable text-to-set generation and multimodal spatial alignment, the proposed framework produces an optical image from the given text, which is then used to construct a modality-shared scene anchor as an explicit structural prior. Specifically, we devise a UNet-based AnchorMap extractor to learn a modality-shared anchor map from modality-specific patterns to constrain structural consistency across modalities. For the multimodal generative network, we implement a conditional diffusion with a structure–attention hybrid sampling (SA-Hybrid) strategy that integrates anchor-based structural control and reference-attention guidance, thus enhancing structural alignment while maintaining modality-specific fidelity. Experiments on benchmarks covering optical, multispectral, synthetic aperture radar (SAR), and digital elevation model (DEM) data show that the proposed method consistently outperforms state-of-the-art baselines in terms of both spatial alignment and modality-specific fidelity. Code will be available at https://github.com/kyh433/Anchor-Diff**

***Index Terms*—AnchorMap extractor, diffusion model, multimodal text-to-set generation, remote sensing, structure–attention hybrid sampling (SA-Hybrid).**

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Yuhan Kang, Honghu Pan, Qiang Liu, and Leyuan Fang are with the School of Artificial Intelligence and Robotics, Hunan University, Changsha 410082, China (e-mail: kyh433@hnu.edu.cn; honghupan@hnu.edu.cn; chiangliu@hnu.edu.cn; fangleyuan@gmail.com).

Hao Liu and Lorenzo Bruzzone are with the Department of Information Engineering and Computer Science, University of Trento, 38123 Trento, Italy (e-mail: hao.liu@unitn.it; lorenzo.bruzzone@unitn.it).

Weiying Xie is with the State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an 710071, China (e-mail: wyxie@xidian.edu.cn).

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## I. Introduction

MULTIMODAL remote sensing imagery, including optical, multispectral, synthetic aperture radar (SAR), and digital elevation model (DEM) data, plays a critical role in Earth observation tasks such as land cover classification [1], [2], environmental monitoring [3], [4], and disaster assessment [5], [6]. The complementary sensing physics and information across modalities enable more robust and comprehensive scene understanding. However, acquiring multiple modalities under consistent spatial and temporal conditions is costly and technically demanding, making well-aligned multimodal datasets scarce. This scarcity limits the development of robust multimodal learning systems and remote sensing foundation models, motivating data-efficient multimodal image generation with strict spatial alignment [4], [7], [8].

Recent advances in generative modeling [9], [10], [11], [12] have significantly improved image generation quality, offering new possibilities for multimodal data generation. In remote sensing, prior studies explored image generation methods for synthesizing realistic, semantically controllable, or modality-translated images [13], [14], [15], [16], [17], [18], [19], [20], [21], [22]. However, these methods are generally developed under single-modality generation settings or predefined source-target modality mappings. As a result, they lack a unified formulation for explicitly modeling the relationships among multiple remote sensing modalities and are difficult to scale to arbitrary multimodal set generation. Although general image-to-image generation methods [23], [24], [25], [26], [27], [28], [29] provide strong translation and conditional control capabilities, multimodal remote sensing generation imposes more stringent requirements: it requires not only fine-grained geometric alignment of multiple geographic targets across modalities, but also high fidelity for each generated modality. Existing image-to-image generation methods usually lack explicit set-level consistency modeling for multimodal outputs. Consequently, structural deviations may accumulate across modalities, resulting in inconsistent shared scene layouts and poor alignment of fine-grained geometry. Moreover, these methods often rely on strong conditional constraints to enforce structural alignment; however, overly strong condition injection may weaken the texture statistics and physical imaging characteristics of the target modality, thereby reducing modality-specific fidelity. Therefore, existing image generation methods remain insufficient for simultaneously achieving strict

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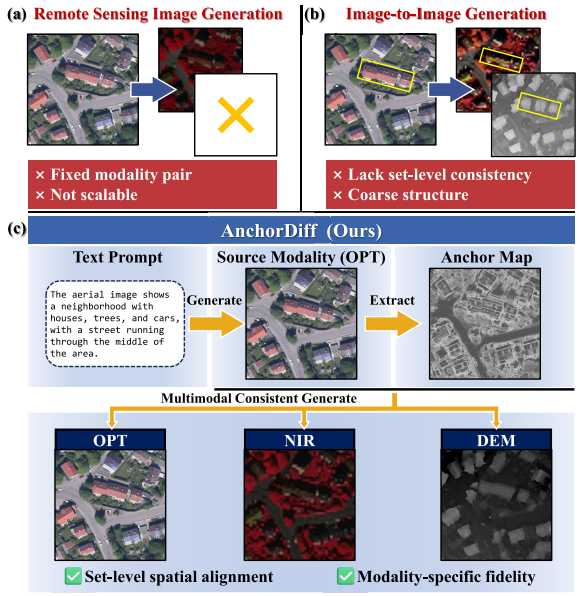



Fig. 1. Motivation of AnchorDiff. (a) Existing remote sensing generation and translation methods are often limited to single-modal synthesis or fixed-pair generation, making them difficult to scale to multimodal set generation. (b) Image-to-image generation methods lack explicit set-level consistency modeling and tend to produce coarse cross-modal structures. (c) AnchorDiff extracts an anchor map from a text-generated optical image as a shared structural condition, enabling spatially aligned multimodal generation with set-level consistency and modality-specific fidelity.

spatial alignment and high modality-specific fidelity in multimodal remote sensing generation [6], [8].

To address these issues, we propose Anchor Diffusion (AnchorDiff), a unified text-to-set framework for generating spatially aligned multimodal remote sensing images. Given a text prompt, AnchorDiff synthesizes a complete set of co-registered modalities, such as Optical, MS, SAR, and DEM, under a shared scene layout, ensuring pixel-level spatial consistency across outputs, as shown in Fig. 1. The key idea is to extract an anchor map that represents the modality-shared scene structure and use it as a common structural reference across modalities, thereby enabling multimodal set generation with explicit set-level consistency. Specifically, we design a UNet-based AnchorMap Extractor to learn modality-shared scene geometry from spatially aligned multimodal samples. During generation, the learned extractor derives an anchor map from the text-generated optical image, which serves as a structural condition for conditional diffusion to constrain the entire multimodal set, leading to fine-grained spatial alignment across modalities. To preserve modality-specific fidelity under strong structural constraints, we further introduce a structure–attention hybrid sampling (SA-Hybrid) strategy for conditional diffusion. It uses target-modality reference attention, obtained under weak structural constraints, to guide strong-constraint sampling, thereby improving structural alignment while preserving modality-specific fidelity. As a result, AnchorDiff achieves spatially aligned multimodal synthesis with high modality-specific fidelity, even under challenging modality gaps.

We evaluate AnchorDiff on multiple benchmarks from optical, multispectral, SAR, and DEM modalities. Extensive experiments show that AnchorDiff consistently outperforms state-of-the-art baselines in terms of spatial alignment and modality-specific fidelity, while maintaining strong text-driven generation quality across diverse scenes.

In summary, our main contributions are as follows.

1) We present **AnchorDiff**, a unified **text-to-set** framework for spatially aligned multimodal remote sensing generation, enabling multimodal set synthesis beyond single-modal generation and specific modality-pair translation.
2) We propose an **AnchorMap Extractor** that learns modality-shared scene geometry from spatially aligned multimodal samples and provides a shared structural prior for set-level spatial alignment across modalities.
3) We introduce a SA-Hybrid strategy for conditional diffusion, which leverages target-modality reference attention to preserve modality-specific fidelity under strong structural constraints.
4) Extensive experiments on Optical/MS/SAR/DEM benchmarks show that AnchorDiff achieves **state-of-the-art** performance on both spatial alignment and modality-specific fidelity across diverse scenes.

## II. Related Work

### A. Remote Sensing Image Generation

Remote sensing image generation aims to synthesize visually plausible samples within specific sensing modalities, such as optical, SAR, or multispectral imagery [12], [30]. It is widely used for data augmentation, rare-scene simulation, and improving the robustness of downstream tasks, including detection and segmentation [12], [30]. Compared with natural-image synthesis, remote sensing generation imposes stricter requirements on both geometric and semantic plausibility. High-resolution remote sensing scenes usually contain abundant multiscale structures, such as road networks, building footprints, and coastlines, whose spatial layouts must remain globally coherent. At the same time, each sensing modality exhibits distinct appearance statistics and degradation patterns, such as speckle noise in SAR imagery [31]. As a result, generative models may produce samples that appear realistic visually while still containing subtle geometric inconsistencies that are unacceptable for remote sensing applications.

Early remote sensing generation methods mainly focused on unconditional generation, where GAN-based models were widely adopted to synthesize visually plausible samples within a single sensing modality [30]. However, these methods often suffer from training instability and limited control over global scene layouts, especially for high-resolution or long-tailed remote sensing data.

In recent years, diffusion models have become a more stable and high-fidelity alternative for remote sensing image generation, including text-conditioned synthesis, controllable generation, inpainting, and super-resolution [12], [13], [14], [15], [16], [32]. A standard diffusion model defines a forward process that gradually perturbs data into Gaussian noise

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{\mathbf{x}_0,\boldsymbol{\epsilon},t}\left[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{c})\|_2^2\right] \tag{1}$$

where $\mathbf{x}_t$ denotes the noisy sample at timestep $t$, and $\boldsymbol{\epsilon}$ denotes Gaussian noise [10]. Based on this paradigm, recent

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diffusion-based methods have been developed to improve the realism, controllability, and semantic alignment of generated remote sensing images. For example, DiffusionSat [13] introduces a diffusion-based generative foundation model tailored to satellite imagery and demonstrates diverse conditional generation capabilities by using remote sensing metadata as guidance. RSDiff [15], CRS-Diff [14], and Text2Earth [16] further explore diffusion-based pipelines to enhance semantic controllability, visual realism, and text-image alignment. Nevertheless, most existing unconditional and text-driven approaches remain essentially single-modal. Although text prompts can provide high-level semantic guidance, they do not explicitly model cross-modality relationships or guarantee pixel-level spatial alignment among different modalities.

Another line of research in remote sensing generation focuses on cross-modal translation, which can help address the lack of explicit modality correspondence in single-modal or text-driven generation. These methods aim to learn mappings between different sensing modalities, such as SAR-to-optical, optical-to-SAR, and related modality conversion tasks [17], [19], [33], [34]. These methods can generate modality-aligned image pairs and preserve coarse scene correspondence to some extent. However, due to the large differences in imaging mechanisms and appearance statistics among optical, multispectral, SAR, and DEM data, existing cross-modal methods usually require targeted designs for specific modality pairs. More importantly, pairwise translation only models the correspondence between one source modality and one target modality, rather than a common scene representation that can be shared across multiple target modalities. Therefore, existing remote sensing image generation methods remain insufficient for arbitrary multimodal set generation, where all generated modalities should follow a shared scene layout.

### B. Image-to-Image Generation

Image-to-image generation provides a general paradigm for conditional image synthesis, where a target image is generated according to a source image, reference image, or spatial control signal. These methods focus on how visual structures can be transferred, edited, or controlled across image domains. Therefore, it offers an important methodological foundation for multimodal generation, especially when spatial correspondence between different outputs is required.

Early image-to-image generation methods are mainly based on generative adversarial networks. Representative paired translation methods, such as pix2pix [23], learn a mapping from a source image $\mathbf{x}$ to a target image $\mathbf{y}$ through conditional adversarial learning

$$\mathcal{L}_{\text{cGAN}} = \mathbb{E}_{\mathbf{x},\mathbf{y}}\left[\log D(\mathbf{x},\mathbf{y})\right] + \mathbb{E}_{\mathbf{x}}\left[\log\left(1 - D(\mathbf{x}, G(\mathbf{x}))\right)\right] \quad (2)$$

where $G$ denotes the generator and $D$ denotes the discriminator. This formulation encourages the generated image $G(\mathbf{x})$ to be indistinguishable from the real target image $\mathbf{y}$ under the source condition $\mathbf{x}$. These GAN-based translation methods demonstrate that image structures can be transferred across domains, but their control mechanism is essentially built upon a source–target mapping. As a result, they are more suitable for single-image translation or bidirectional domain transfer, rather than generating a coordinated set of multiple outputs.

Beyond pairwise translation, multidomain and shared-representation methods extend image translation by modeling reusable information across domains. Multidomain approaches, such as StarGAN [35], use a unified generator to translate an input image toward different target domains, while shared-latent methods, such as UNIT [36] and MUNIT [37], represent cross-domain content through shared or domain-invariant latent features. These formulations improve scalability across domains and capture domain-common content more explicitly than pairwise mappings. However, they still generate individual target-domain images separately and do not explicitly constrain multiple outputs to share identical fine-grained scene geometry as a coordinated set.

Diffusion-based image-to-image generation methods further improve generation quality by combining strong generative priors with conditional denoising. Methods such as Palette [25], SDEdit [26], Pix2Pix-Zero [27], and Brownian-bridge diffusion models [28] formulate image-to-image generation as a stochastic denoising or editing process conditioned on a source image or an intermediate image representation. Compared with GAN-based translation, diffusion-based methods provide a more flexible generation process and can better preserve image realism under diverse conditional settings. More recently, controllable diffusion methods, represented by ControlNet [29], introduce explicit spatial control signals into pretrained diffusion models. Given a spatial condition $\mathbf{s}$, such as an edge map, depth map, segmentation map, pose map, or other structural representation, the denoising network predicts the injected noise under both the noisy target image and the control signal

$$\mathcal{L}_{\text{cond}} = \mathbb{E}_{\mathbf{y}_0,\boldsymbol{\epsilon},t}\left[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{y}_t, t, \mathbf{s})\|_2^2\right] \quad (3)$$

where $\mathbf{y}_t$ denotes the noisy target image at timestep $t$, $\boldsymbol{\epsilon}$ denotes Gaussian noise, and $\mathbf{s}$ denotes the spatial control signal. This conditional formulation significantly improves structural controllability, since the generation process can be constrained by explicit geometric or semantic layouts.

However, existing image-to-image and controllable diffusion methods usually control each generation process independently and lack explicit set-level consistency modeling for multiple outputs. A common structural condition, such as an edge, depth, or segmentation map, can in principle be reused across different target domains, but it is typically externally provided or derived from a single input and applied independently to each generation process. Consequently, sharing the same control signal does not explicitly model a modality-shared scene representation or guarantee identical fine-grained geometry across all generated outputs. When extended to multimodal generation, such independent conditional generation may introduce local shape variations, boundary shifts, and inconsistent fine-grained layouts across different outputs, even when they are guided by similar or shared spatial conditions.

Another limitation is the trade-off between structural alignment and modality-specific fidelity. Image-to-image generation

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methods often rely on strong conditional constraints to enforce structural consistency. While stronger control can improve alignment with the given condition, overly strong condition injection may over-constrain the denoising trajectory and weaken the target domain's texture statistics, local appearance patterns, and physical characteristics. This issue becomes more critical under large modality gaps, where different outputs should share the same underlying structure while still exhibiting distinct domain-specific responses. Therefore, multimodal remote sensing image generation requires not only set-level spatial consistency across modalities, but also modality-specific fidelity under strong structural constraints.

Different from pairwise translation, shared-latent generation, and independently applied structural control, AnchorDiff explicitly learns a modality-shared anchor map from aligned multimodal observations and distills this set-level structure for single-source inference. The recovered anchor is then shared across target modalities to constrain their spatial layouts, rather than generating each modality from an independently inferred scene representation. Moreover, the proposed SA-Hybrid balances this common structural constraint with modality-specific fidelity under large cross-modal differences.

## III. Methodology

### A. Overall Framework

In multimodal text-to-set remote sensing image generation, a text prompt serves as the primary input, and the model is required to generate a set of remote sensing images depicting the same scene under the specified modalities. As illustrated in Fig. 2, AnchorDiff organizes this task in a source–anchor–target generation pipeline. It first synthesizes a source-modality image from the text prompt using a text-conditioned diffusion backbone to instantiate the scene content and initial spatial layout. The source image is then fed into the AnchorMap Extractor to estimate a modality-shared anchor map, which highlights structural regions that are expected to remain stable across heterogeneous sensing modalities. For each target modality, AnchorDiff combines the text condition, the modality condition, the source image, and the anchor map to perform target-modality synthesis. In this process, the anchor map is used to construct masked structural guidance, so that source-derived structural cues are selectively injected into cross-modal stable regions rather than transferred as modality-specific appearance. To further improve the fidelity of generated target modalities during conditional generation, AnchorDiff adopts SA-Hybrid, which uses a weakly guided reference pass and a strongly guided refinement pass to jointly preserve modality-specific attention responses and enhance anchor-based spatial consistency. Sections III-B and III-C detail these two key components: AnchorMap Extractor and SA-Hybrid.

### B. AnchorMap Extractor

To preserve geometric consistency across modalities, we introduce an intermediate structural representation, termed the anchor map, and use it as the structural condition for target-modality generation. Rather than representing raw modality-dependent edges, the anchor map is defined as a modality-shared scene anchor that captures cross-modal stable scene structure while suppressing modality-specific texture and appearance responses. It therefore serves as an explicit structural prior that preserves scene layout, object contours, and spatially stable geometric patterns shared across heterogeneous sensing modalities. Motivated by this observation, we design an anchor map extractor to learn such a cross-modal stable representation from aligned multimodal observations, so that the shared scene structure can be reliably recovered and used as the structural condition for subsequent target-modality synthesis.

The AnchorMap Extractor is implemented as a lightweight shared-region prediction network. Given aligned multimodal observations, each modality is first processed by an independent convolutional encoder, producing three-scale features. We use modality-specific encoders because different sensing modalities usually exhibit distinct low-level statistics. In the multimodal branch, shallow and middle-level features are fused by cross-modal averaging, while the deepest features are fused by an attention bottleneck. Specifically, the deep features from all modalities are concatenated and passed through a small convolutional context module to predict modality-wise spatial attention weights, which are normalized by a softmax operation and used to obtain the fused deep feature. A shared U-Net-like decoder then upsamples the fused features with skip connections and predicts a single-channel mask through a sigmoid layer.

Formally, let $\mathcal{X} = \{\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n\}$ denote aligned observations of the same scene from $n$ modalities. For each modality $\mathbf{X}_i$, a modality-specific encoder extracts structural features and predicts a unimodal anchor map $\mathbf{M}_i \in [0,1]^{H\times W}$. In parallel, the modality-specific features are fused through a cross-modal interaction module to produce a fused feature representation, from which a multimodal anchor prediction $\mathbf{M}_{\text{multi}} \in [0,1]^{H\times W}$ is obtained. Here, $\mathbf{M}_i$ denotes the anchor map estimated from a single modality, while $\mathbf{M}_{\text{multi}}$ denotes the anchor map inferred from all available aligned modalities during training. In this way, the fused branch provides a stronger estimate of the shared structural content, while the unimodal branches learn to recover the same scene anchor from individual modalities.

Since no ground-truth annotation is available for the anchor map, its supervision is derived from cross-modal structural consistency among aligned observations. Specifically, for each modality $\mathbf{X}_i$, we compute a normalized gradient magnitude map $\mathbf{G}_i$. Although heterogeneous modalities exhibit different radiometric responses and contrast characteristics, major scene boundaries are often spatially co-located in aligned observations. Such stable structures typically include road layouts, building footprints, field parcels, water–land interfaces, vegetation boundaries, and major terrain discontinuities. Gradient magnitude captures such local discontinuities without depending on response polarity, while normalization places different modalities on a comparable response scale. We then aggregate these responses across modalities to form a cross-modal structure prior

$$\mathbf{P}_{\text{struct}} = \text{Aggregate}\,(\mathbf{G}_1, \mathbf{G}_2, \ldots, \mathbf{G}_n) \tag{4}$$

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where Aggregate(·) denotes a cross-modal aggregation operator, instantiated as median aggregation in our implementation. Compared with maximum or mean aggregation, the median is less affected by strong modality-specific responses, while avoiding the excessive suppression of valid structures caused by minimum aggregation. Because the inputs are spatially aligned, locations that consistently exhibit strong structural responses across modalities are more likely to correspond to stable scene geometry, whereas modality-specific texture fluctuations tend to be suppressed through aggregation. Thus, $\mathbf{P}_{\text{struct}}$ is used as a weak structural prior rather than a ground-truth anchor map annotation.

Based on the above formulation, the extractor is optimized with three complementary objectives. First, we adopt a teacher–student distillation loss to transfer the shared structural knowledge learned by the multimodal fused branch to each unimodal branch. Specifically, the fused prediction $\mathbf{M}_{\text{multi}}$ is treated as a stronger teacher signal during training, and each unimodal prediction $\mathbf{M}_i$ is encouraged to approximate it

$$\mathcal{L}_{\text{distill}} = \frac{1}{n}\sum_{i=1}^{n} \left\| \mathbf{M}_i - \text{sg}\left(\mathbf{M}_{\text{multi}}\right) \right\|_1 \tag{5}$$

where sg(·) denotes the stop-gradient operation. This loss follows a standard distillation form, but is adapted here to enable single-modality inference of the modality-shared scene anchor.

Second, we introduce a prior-consistency objective to align the fused anchor prediction with the cross-modal structure prior

$$\mathcal{L}_{\text{prior}} = 1 - \text{Corr}\left(\mathbf{M}_{\text{multi}}, \mathbf{P}_{\text{struct}}\right) \tag{6}$$

where $\text{Corr}(\cdot,\cdot)$ denotes the sample-wise correlation between the predicted anchor map and the aggregated structural prior. Unlike the distillation and total-variation terms, which are standard regularization forms, $\mathcal{L}_{\text{prior}}$ is task-specific in that it is built upon the proposed cross-modal structural aggregation prior and serves as the key weak supervisory signal for anchor learning without manual annotation.

Third, to suppress isolated noisy responses and improve spatial coherence, we impose a total variation (TV) regularization on the fused anchor prediction

$$\mathcal{L}_{\text{tv}} = \|\nabla_u \mathbf{M}_{\text{multi}}\|_1 + \|\nabla_v \mathbf{M}_{\text{multi}}\|_1 \tag{7}$$

where $\nabla_u$ and $\nabla_v$ denote finite differences along the horizontal and vertical directions, respectively, and the $\ell_1$ norm is implemented as the mean absolute difference in our implementation. This regularizer is a standard smoothness prior that encourages the anchor map to form spatially coherent structural responses rather than fragmented local activations.

The overall training objective is therefore written as

$$\mathcal{L}_{\text{extractor}} = \lambda_d \mathcal{L}_{\text{distill}} + \lambda_p \mathcal{L}_{\text{prior}} + \lambda_{tv} \mathcal{L}_{\text{tv}}. \tag{8}$$

In our experiments, the weighting coefficients are empirically set to $\lambda_d = \lambda_p = \lambda_{tv} = 1$. In summary, $\mathcal{L}_{\text{distill}}$ ensures that the shared scene anchor remains recoverable from a single modality at inference time, $\mathcal{L}_{\text{prior}}$ injects cross-modal structural supervision without requiring manual labels, and $\mathcal{L}_{\text{tv}}$ improves the spatial coherence of the learned anchor representation. During training, the multimodal branch learns the shared scene anchor from aggregated structural evidence across aligned modalities, while each unimodal branch is distilled to approximate this fused prediction.

During inference, given a single source-modality image $\mathbf{I}_{\text{src}}$, the trained AnchorMap Extractor predicts the corresponding anchor map as

$$\mathbf{M}_{\text{src}} = \mathcal{E}\left(\mathbf{I}_{\text{src}}\right) \tag{9}$$

where $\mathcal{E}(\cdot)$ denotes the trained extractor used at inference time. In Section III-C, we further incorporate this anchor-based structural prior into the proposed SA-Hybrid scheme to guide the diffusion process toward better scene layout preservation and local spatial alignment across modalities.

### C. Structure–Attention Hybrid Sampling

Given the anchor map predicted from the source modality, the remaining challenge is how to incorporate such a modality-shared structural prior into target-modality diffusion while preserving both structural consistency and target-specific fidelity. Although anchor-based structural control can effectively improve cross-modal alignment, overly strong structural guidance may distort the characteristic appearance and sensing properties of the target modality. To address this issue, we propose a SA-Hybrid strategy that combines anchor-based structural control with reference-attention guidance during target-modality denoising. The key idea is to preserve a target-faithful attention reference under milder structural constraints and then use it to regularize denoising under stronger anchor-based control. In this way, the proposed strategy improves structural alignment under the shared scene anchor while maintaining modality-specific fidelity.

We first generate the source-modality image from the input text prompt using a source-modality diffusion model, yielding a source image $\mathbf{I}_{\text{src}}$. The proposed AnchorMap Extractor is then applied to $\mathbf{I}_{\text{src}}$ to obtain the corresponding anchor map $\mathbf{M}_{\text{src}}$. In target-modality denoising, the source image provides scene-level structural context, while the anchor map determines where this structural condition should be emphasized. Since the source-derived structural residual may contain both modality-shared geometry and source-specific appearance responses, directly injecting the full residual can introduce undesirable source-modality characteristics into the target generation. Let $\mathbf{R}_t^{(l)}$ denote the structural residual injected into the $l$th denoising block at timestep $t$. We resize the anchor map to the corresponding feature resolution and use the standardized map as a spatial mask

$$\tilde{\mathbf{R}}_t^{(l)} = \mathbf{M}_{\text{src}}^{(l)} \odot \mathbf{R}_t^{(l)} \tag{10}$$

where $\mathbf{M}_{\text{src}}^{(l)}$ denotes the anchor map resized to the spatial resolution of the $l$th feature map, and $\odot$ denotes element-wise multiplication. The masked residual $\tilde{\mathbf{R}}_t^{(l)}$ is then injected into the target denoising network as the structural guidance. This masking emphasizes source guidance in cross-modal stable regions while attenuating source-specific responses elsewhere. It only modulates the conditional residual and does not alter the noisy latent or the forward diffusion process.

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To balance modality-specific fidelity and structural consistency, target-modality sampling is performed through two complete conditional DDIM passes, both initialized from the same Gaussian latent $\mathbf{z}_T$ and using the same timestep schedule. In the first reference pass, we apply a relatively mild structural guidance scale and perform structurally guided denoising to establish a target-faithful attention reference. Since the structural constraint is not yet dominant in this pass, the resulting denoising trajectory better reflects the intrinsic style and sensing characteristics of the target modality. During this process, we cache the cross-attention maps produced by the target denoising network at each timestep and layer, and use them as a fidelity-aware reference for the subsequent refinement pass. Let $\mathbf{A}_{t,l}^{\text{ref}}$ denote the cross-attention maps at timestep $t$ and layer $l$ during the reference pass. These attention maps are stored as

$$\mathbf{A}_{t,l}^{\text{ref}} = \text{Attention}_{\text{tar}}\left(\mathbf{x}_t^{(l)}\right) \tag{11}$$

where $x_t^{(l)}$ is the input feature at timestep $t$ and layer $l$, and $\text{Attention}_{\text{tar}}$ refers to the cross-attention function used to compute the attention maps during the first pass of denoising. These cached attention maps are then used in the second pass to refine the target-modality generation, preserving target-specific fidelity while maintaining strong structural consistency. Only the cached attention maps are retained as reference information; no intermediate or terminal latent state from the reference pass is propagated to the refinement pass.

In the second pass, we restart denoising from the same initial latent code and apply a stronger structural guidance scale to enforce more explicit structural consistency with the shared scene anchor. At each timestep $t$, the current latent is first processed under the same masked structural guidance, so that geometrically stable shared regions are more strongly constrained. However, because excessively strong structural guidance may distort target-modality characteristics, we additionally regularize the current cross-attention maps using the cached reference attention maps obtained in the first pass. Let $\mathbf{A}_{t,l}^{\text{cur}}$ denote the current cross-attention maps at timestep $t$ and layer $l$ in the refinement pass. The attention regularization objective is defined as

$$\mathcal{L}_{\text{attn}}^{(t)} = \sum_l \left\| \mathbf{A}_{t,l}^{\text{cur}} - \text{sg}\left(\mathbf{A}_{t,l}^{\text{ref}}\right) \right\|_2^2 \tag{12}$$

where $\text{sg}(\cdot)$ denotes the stop-gradient operation. The current latent is updated according to the gradient of $\mathcal{L}_{\text{attn}}^{(t)}$, and the refined latent is then used for the subsequent scheduler step under the stronger structural guidance.

The proposed hybrid mechanism benefits from the complementary roles of these two passes. The structural guidance branch ensures that the generated target modality remains aligned with the shared scene structure extracted from the source modality, especially in cross-modal stable regions emphasized by the anchor map. The attention branch, in contrast, preserves target-modality fidelity by recording the attention organization formed under milder structural guidance and pulling the strongly controlled denoising trajectory back toward this target-specific configuration. As a result, the proposed sampling scheme explicitly balances structural consistency and modality-specific fidelity: the reference pass establishes a fidelity-aware target trajectory, while the refinement pass strengthens structural alignment and simultaneously prevents target-modality characteristics from being excessively distorted. The complete inference procedure of the proposed framework is summarized in Algorithm 1.

**Algorithm 1** Inference Algorithm of AnchorDiff

**Require:** Text prompt $P$, negative prompt $P_{\text{neg}}$, target modality set $\mathcal{M} = \{m_1, \ldots, m_K\}$, denoising steps $T$, weak and strong guidance scales $\gamma_w$ and $\gamma_s$.
**Ensure:** Multimodal image set $\mathcal{I}_{\text{multi}} = \{\mathbf{I}^{(m)}\}_{m\in\mathcal{M}}$.
1. Encode $(P, P_{\text{neg}})$ into text condition embeddings.
2. Generate the source image $\mathbf{I}_{\text{src}}$ from the text condition.
3. Predict the anchor map $\mathbf{M}_{\text{src}}$ as in Eq. (9).
4. Use $\mathbf{I}_{\text{src}}$ as the structural condition.
5. Construct masked structural guidance from $\mathbf{M}_{\text{src}}$ as in Eq. (10).
**for** each target modality $m \in \mathcal{M}$ **do**
6. Sample an initial noisy latent $\mathbf{z}_T^{(m)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
6.1 Starting from $\mathbf{z}_T^{(m)}$, perform a complete DDIM reverse trajectory under weak guidance $\gamma_w$.
6.2 Cache only the attention maps $\mathbf{A}_{t,l}^{\text{ref}}$ as in Eq. (11).
6.3 Restart from the same $\mathbf{z}_T^{(m)}$ and perform a complete DDIM reverse trajectory with the same timesteps and scheduler under strong guidance $\gamma_s$.
6.4 Apply the attention-guidance correction before each DDIM update according to Eq. (12).
7. Decode the refined latent to obtain $\mathbf{I}^{(m)}$.
**end for**
8. Collect $\{\mathbf{I}^{(m)}\}_{m\in\mathcal{M}}$ as $\mathcal{I}_{\text{multi}}$.
**return:** $\mathcal{I}_{\text{multi}}$.

## IV. Experiments

### A. Experimental Setup

*1) Datasets:* We evaluate the proposed method on four publicly available multimodal remote sensing datasets, namely **FUSAR-MaP** [38], **Hunan-Dataset** [39], **DFC2019** [40], and **MDAS** [41]. These datasets cover diverse multimodal combinations, including optical (OPT), near-infrared (NIR), SAR, and DEM data, and therefore provide a suitable benchmark for text-to-aligned multimodal generation.

Specifically, FUSAR-MaP contains paired OPT and SAR data collected from eight regions across six provinces in China, both with a spatial resolution of 1.02 m. Hunan-Dataset covers Hunan Province, China, and includes OPT, NIR, and SAR images with a spatial resolution of 10 m. DFC2019 contains urban scenes from Jacksonville, Florida, and Omaha, Nebraska, USA, including OPT, NIR, and DEM modalities, with spatial resolutions of 0.35, 1.3, and 0.8 m, respectively. MDAS covers Augsburg, Germany, and provides OPT, NIR, and DEM data with approximate spatial resolutions of 0.2, 2.0, and 0.67 m, respectively.

For all datasets, the aligned multimodal images are cropped into 512 × 512 tiles. The data are then divided into training and testing sets with a ratio of 80:20. Following the

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TABLE I
SUMMARY OF THE MULTIMODAL DATASETS USED IN OUR EXPERIMENTS. TRAIN AND TEST DENOTE THE NUMBERS OF ALIGNED MULTIMODAL SCENE TILES

| Dataset | OPT | NIR | SAR | DEM | Train | Test |
|---|---|---|---|---|---|---|
| FUSAR-MaP [38] | ✓ | □ | ✓ | □ | 1952 | 488 |
| Hunan-Dataset [39] | ✓ | ✓ | ✓ | □ | 400 | 100 |
| DFC2019 [40] | ✓ | ✓ | □ | ✓ | 345 | 86 |
| MDAS [41] | ✓ | ✓ | □ | ✓ | 5040 | 1260 |

caption-based protocol commonly adopted in remote-sensing text-to-image generation and vision-language dataset construction [15], [42], we use the optical image of each multimodal tile as the semantic reference and employ InstructBLIP-Vicuna-7B to generate a scene-level caption describing its principal geographic objects and scene content. Since the corresponding OPT, NIR, SAR, and DEM observations are spatially aligned representations of the same scene, the caption generated from the optical image is shared by all available modalities in that set as their common textual condition. This protocol ensures consistent text–scene correspondence and enables evaluation of text-conditioned multimodal generation on unseen test scenes across different modality combinations. The main statistics of the four datasets are summarized in Table I.

*2) Evaluation Metrics:* We evaluate the proposed method using four metrics, namely **FID** [43], **KID** [44], **CLIP-Score** [45], and **NMI** [46], from three complementary perspectives: modality-specific fidelity, text–image semantic consistency, and cross-modal spatial alignment.

Specifically, FID and KID are used to assess the fidelity of each generated modality by measuring its distribution discrepancy from real test images, where lower values indicate better realism. CLIP-Score is adopted to evaluate the semantic consistency between each generated modality and its input caption, where higher values indicate better text-conditioned generation quality. NMI is used to quantify the shared structural information between generated modality pairs within the same scene; higher values indicate stronger cross-modal consistency and spatial alignment.

In addition to generation quality metrics, we further evaluate the extracted anchor maps using three complementary metrics, including **Edge Correlation** (Edge Corr), **MAE**, and **TV** [47]. Different from conventional reconstruction evaluation, the quality of an anchor map is determined primarily by whether it preserves meaningful cross-modal structural information. Therefore, Edge Corr is considered the key structural indicator, where a higher value indicates stronger correspondence between the extracted anchor map and the cross-modal structural response. Meanwhile, MAE measures the consistency between the multimodal anchor map $M_{\text{multi}}$ and the corresponding unimodal predictions $M_i$, where a lower value indicates better agreement. TV [47] evaluates the spatial variation of the predicted anchor map, where a lower value indicates fewer isolated responses and improved spatial coherence.

*3) Implementation Details:* Our framework is built upon the pre-trained Stable Diffusion 1.5 [11]. In the current implementation, OPT is adopted as the source modality for all experiments. This choice is based on the availability of OPT across all four benchmark datasets and its direct semantic correspondence with natural-language descriptions and pretrained text-to-image priors. For each dataset, the diffusion backbone is fine-tuned on the corresponding multimodal training data using the standard noise-prediction objective, conditioned on the text prompt and the corresponding modality condition. The AnchorMap Extractor is trained separately using aligned multimodal images. The resulting diffusion weights are further used to initialize the ControlNet-based baseline to ensure a fair comparison under a consistent generative prior. Unless otherwise specified, all generated samples have a resolution of 512 $\times$ 512. The proposed SA-Hybrid is applied only at inference and requires no additional diffusion training. During inference, the number of DDIM sampling steps is set to 50, and the classifier-free guidance scale is set to 7.5. All models are trained and evaluated on a single NVIDIA GeForce RTX 3090 GPU.

### B. Quantitative and Qualitative Comparisons

*1) Quantitative Comparison:* Table II reports the quantitative comparison between AnchorDiff and four representative baselines: DiffusionSat [13], CycleGAN [24], Pix2Pix-Zero [27], and CRS-Diff [14], across four datasets. The compared methods cover single-modal remote sensing generation and cross-domain translation, providing a comparison of different translation paradigms.

To evaluate AnchorDiff's performance in comparison to single-modal remote sensing generation methods, we contrast it with DiffusionSat [13]. While DiffusionSat [13] demonstrates reasonable image fidelity for individual modalities, its generation process is primarily conditioned on text prompts, without enforcing a shared spatial structure. This limitation restricts its ability to produce cross-modal consistency. The results highlight that single-modal, text-conditioned generation alone is insufficient for generating spatially aligned multimodal outputs. In contrast, AnchorDiff significantly outperforms DiffusionSat [13] in terms of cross-modal consistency, as evidenced by its superior performance in metrics like NMI and CLIP-Score.

In the realm of cross-domain image translation, we compare AnchorDiff with CycleGAN [24], Pix2Pix-Zero [27], and CRS-Diff [14]. CycleGAN [24] relies on cycle consistency to preserve coarse content but lacks an explicit mechanism for modeling the shared geometry among heterogeneous remote sensing modalities, leading to significant performance degradation, particularly when there are large modality gaps. While Pix2Pix-Zero [27] introduces diffusion priors and outperforms traditional GAN-based methods in some cases, it still follows a generic translation paradigm that does not model the shared structures across remote sensing modalities. CRS-Diff [14] benefits from controllable diffusion and achieves competitive performance on some alignment-related and semantic metrics, but its image fidelity varies across datasets. These comparisons emphasize that cross-domain image translation methods struggle to capture the modality-shared structures necessary for effective multimodal remote sensing generation.

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TABLE II

QUANTITATIVE COMPARISON ACROSS FOUR DATASETS. ONLY MEAN VALUES ARE REPORTED. BOLD VALUES INDICATE THE BEST RESULTS, AND "-" INDICATES THAT THE CORRESPONDING MODALITY OR METRIC IS UNAVAILABLE OR OMITTED

| Method | Modality | FUSAR-MaP | | | | Hunan-DataSet | | | | DFC2019 | | | | MDAS | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | FID↓ | KID↓ | NMI↑ | CLIP↑ | FID↓ | KID↓ | NMI↑ | CLIP↑ | FID↓ | KID↓ | NMI↑ | CLIP↑ | FID↓ | KID↓ | NMI↑ | CLIP↑ |
| | OPT | 111.273 | 0.0316 | - | 0.2754 | 224.560 | 0.0329 | - | 0.2714 | 196.199 | 0.0146 | - | 0.3007 | 56.548 | 0.0132 | - | 0.2957 |
| DiffusionSat [13] | NIR | - | - | - | - | 239.565 | 0.0480 | 0.1230 | 0.2589 | 217.239 | 0.0402 | 0.1244 | 0.2651 | 82.670 | 0.0260 | 0.1238 | 0.2014 |
| | SAR | 98.734 | 0.0626 | 0.1209 | 0.2342 | 218.078 | 0.0524 | 0.1176 | 0.2426 | - | - | - | - | - | - | - | - |
| | DEM | - | - | - | - | - | - | - | - | 187.026 | 0.0474 | 0.1109 | 0.1856 | 68.930 | 0.0295 | 0.1089 | 0.2176 |
| CycleGAN [24] | NIR | - | - | - | - | 242.170 | 0.0487 | 0.2250 | 0.2560 | 222.029 | 0.0365 | 0.1902 | 0.2353 | 267.398 | 0.2547 | 0.1280 | 0.1437 |
| | SAR | 192.640 | 0.1588 | 0.1340 | 0.2402 | 223.157 | 0.0612 | 0.1457 | 0.2359 | - | - | - | - | - | - | - | - |
| | DEM | - | - | - | - | - | - | - | - | 223.148 | 0.0627 | 0.1179 | 0.1917 | 207.235 | 0.1644 | 0.1499 | 0.1838 |
| Pix2Pix-Zero [27] | NIR | - | - | - | - | 243.870 | 0.0469 | 0.1337 | 0.2523 | 227.603 | 0.0382 | 0.1322 | 0.2660 | 90.561 | 0.0460 | 0.1370 | 0.1907 |
| | SAR | 133.296 | 0.0989 | 0.1333 | 0.2383 | 221.489 | 0.0537 | 0.1320 | 0.2432 | - | - | - | - | - | - | - | - |
| | DEM | - | - | - | - | - | - | - | - | 182.057 | 0.0367 | 0.1212 | 0.1866 | 66.038 | 0.0296 | 0.1303 | 0.2164 |
| CRS-Diff [14] | NIR | - | - | - | - | 258.480 | 0.0610 | 0.2374 | 0.2561 | 220.618 | 0.0368 | 0.2061 | 0.2789 | 96.318 | 0.0569 | 0.1601 | 0.2089 |
| | SAR | 151.665 | 0.1240 | 0.1399 | 0.2383 | 248.771 | 0.0562 | 0.1463 | 0.2556 | - | - | - | - | - | - | - | - |
| | DEM | - | - | - | - | - | - | - | - | 188.879 | 0.0318 | 0.1223 | 0.1847 | 118.220 | 0.0751 | 0.1531 | 0.2209 |
| AnchorDiff | NIR | - | - | - | - | **238.019** | **0.0413** | **0.2393** | **0.2597** | **207.044** | **0.0323** | **0.2175** | **0.2803** | **59.856** | **0.0205** | **0.1685** | **0.2107** |
| | SAR | **86.095** | **0.0508** | **0.1419** | **0.2429** | **216.619** | **0.0513** | **0.1509** | **0.2599** | - | - | - | - | - | - | - | - |
| | DEM | - | - | - | - | - | - | - | - | **175.132** | **0.0200** | **0.1239** | **0.2034** | **62.149** | **0.0282** | **0.1591** | **0.2288** |

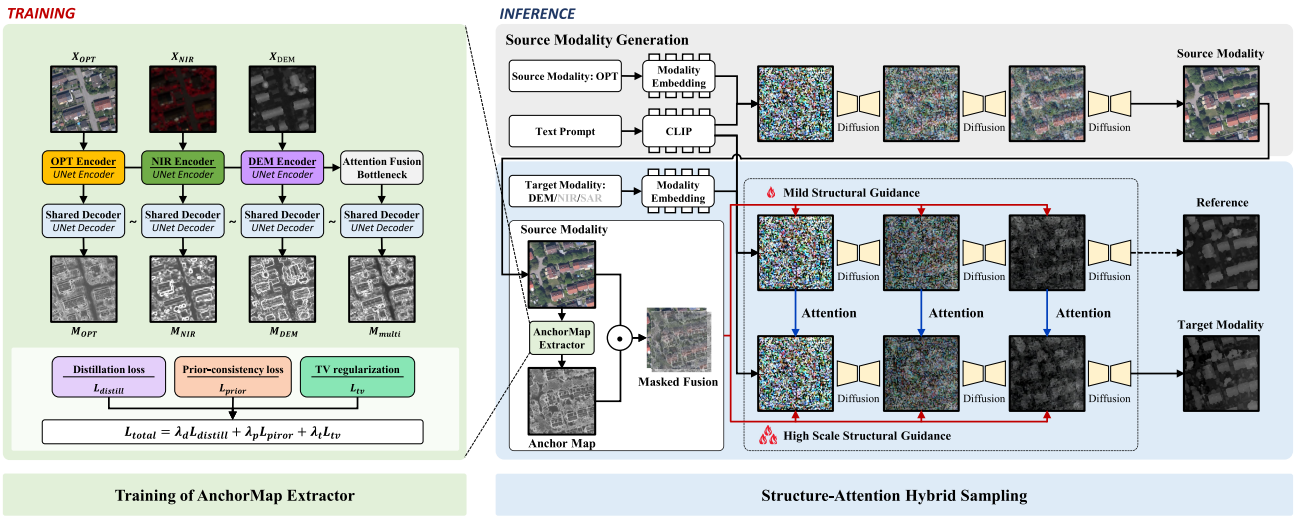



Fig. 2. Overview of AnchorDiff. (Left) The proposed AnchorMap Extractor learns a modality-shared scene anchor from aligned multimodal observations and supervises anchor prediction with distillation, prior-consistency, and total-variation losses. (Right) Given a text prompt, AnchorDiff first generates a source-modality image, and then applies the proposed SA-Hybrid to produce spatially aligned target modalities. Specifically, the extracted anchor map is used as masked structural guidance, while a reference pass with mild guidance and a refinement pass with stronger guidance jointly preserve structural alignment and modality-specific fidelity.

Overall, AnchorDiff achieves the most balanced performance across different datasets and modalities. It consistently improves fidelity-related metrics such as FID and KID, while also maintaining competitive or superior cross-modal alignment and semantic consistency measured by NMI and CLIP-Score.

*2) Qualitative Comparison:* Fig. 3 further presents qualitative comparisons on representative scenes from the four datasets. The selected examples contain diverse land-cover and land-object patterns, including agricultural fields, roads, buildings, water bodies, vegetation, and elevation-related structures. Consistent with the quantitative results, the baseline methods can generate visually plausible results in some individual modalities, but they often suffer from artifacts, over-smoothed details, unrealistic modality-specific textures, or structural drift. More importantly, their generated modalities frequently exhibit inconsistent spatial layouts, such as shifted roads, distorted building boundaries, mismatched field parcels, and inaccurate water or vegetation regions. Specifically, CycleGAN [24] tends to preserve only coarse scene layouts and may introduce severe artifacts when the modality gap is large. Pix2Pix-Zero [27] and CRS-Diff [14] can improve local appearance in some cases, but they still show unstable structural correspondence across NIR, SAR, and DEM outputs. In contrast, AnchorDiff produces results that are more consistent with the GT in both global layout and local structures. The generated roads, building boundaries, field parcels, water regions, and terrain patterns are better aligned across modalities, while modality-specific characteristics are also more faithfully preserved. These visual comparisons indicate that AnchorDiff does not merely improve image realism, but also enhances cross-modal structural coherence.

*3) Structural Consistency Visualization:* While Fig. 3 provides visual evidence from generated multimodal images, Fig. 4 offers a more direct diagnostic visualization of cross-modal structural consistency from the perspective of anchor map discrepancy. To further evaluate whether the source-derived structural anchor is consistent with the shared

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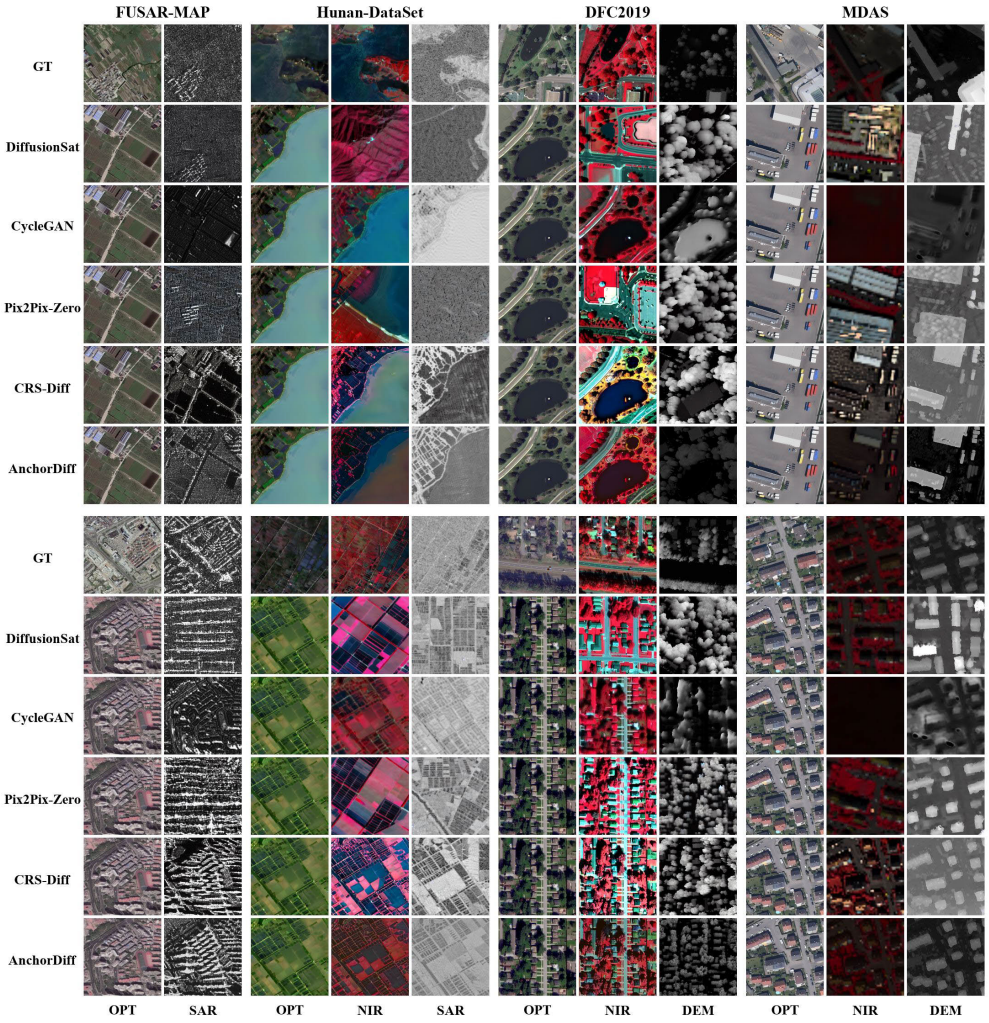



Fig. 3. Qualitative comparison across four datasets and multiple modalities. The results include representative land-cover structures such as roads, buildings, vegetation, water bodies, and agricultural fields. GT denotes the aligned ground-truth multimodal images.

structure of the generated multimodal set, we visualize the discrepancy between $M_{\text{OPT}}$, extracted from the generated optical image, and $M_{\text{multi}}$, inferred from all generated modalities by the AnchorMap Extractor. Specifically, we compute their absolute difference as the structural discrepancy map, where darker regions indicate stronger consistency between the optical anchor and the multimodal shared anchor. As shown in Fig. 4, the baseline methods exhibit evident discrepancy responses around roads, building contours, water boundaries, and dense urban patterns, revealing that the optical-side structure is not well aligned with the shared structural patterns of the generated multimodal outputs. By contrast, AnchorDiff produces darker discrepancy maps with fewer high-response regions across different datasets and scenes. This indicates that the learned anchor map provides a cross-modal stable structural representation, as the anchor extracted from the source modality remains close to the set-level shared anchor rather than being dominated by modality-specific texture or local appearance responses. These results provide direct visual evidence that AnchorDiff better preserves modality-shared scene structures and maintains stronger set-level spatial consistency across generated modalities.

Overall, the quantitative and qualitative results consistently validate the central design of AnchorDiff. By modeling a modality-shared anchor, the proposed approach effectively balances image fidelity, semantic consistency, and spatial alignment, making it particularly suitable for unified text-to-aligned multimodal remote sensing generation tasks.

### C. Ablation Study

We conduct ablation studies at two levels to investigate the effectiveness of the proposed components in AnchorDiff. First, we analyze the training objective of the AnchorMap Extractor by evaluating different combinations of loss functions. Then, using the resulting AnchorMap Extractor, we systematically evaluate the key components of the complete AnchorDiff framework, including the structural condition, Masked Fusion, and SA-Hybrid.

*1) Ablation of AnchorMap Extractor Training Objective:* To further investigate the contribution of each loss term in the AnchorMap Extractor, we first perform a dedicated ablation study on different loss combinations, including $L_{distill}$, $L_{distill} + L_{prior}$, and $L_{distill} + L_{prior} + L_{tv}$. As shown in Fig. 5,

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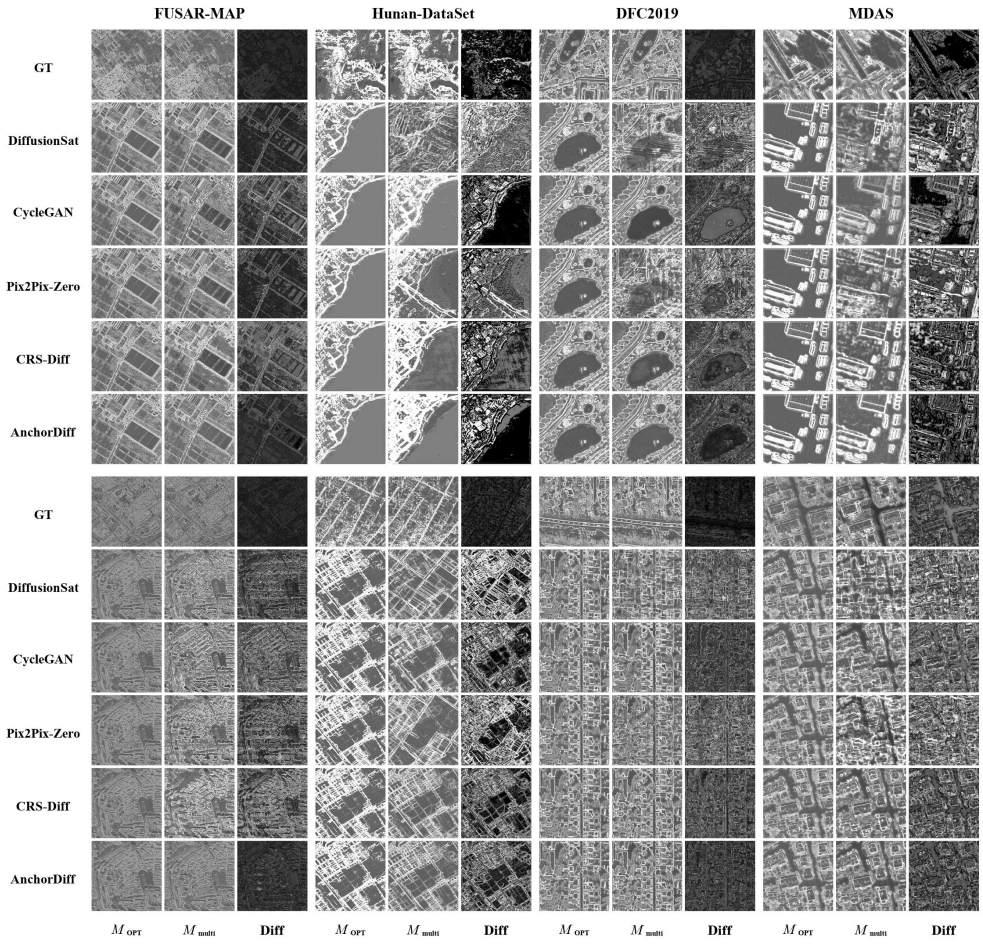



Fig. 4. Qualitative comparison of cross-modal structural consistency. The difference between $M_{\text{OPT}}$ and $M_{\text{multi}}$ is visualized, where darker regions indicate stronger structural consistency across generated modalities.

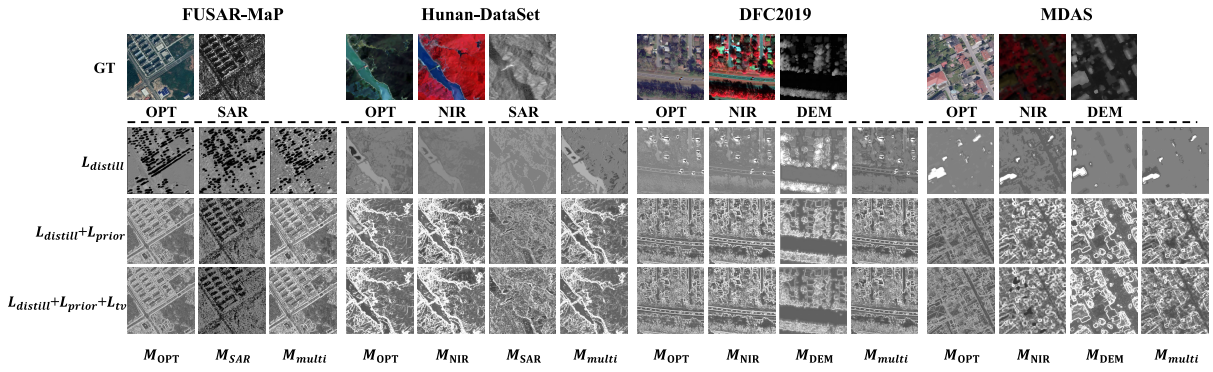



Fig. 5. Qualitative comparison of extracted anchor maps under different loss configurations of the AnchorMap Extractor.

when only $L_{distill}$ is employed, the extracted unimodal maps can effectively approximate the multimodal prediction $M_{\text{multi}}$, indicating that the distillation objective successfully transfers the shared representation from the multimodal branch to individual modalities. However, these anchor maps exhibit weak correspondence with salient scene structures, where important object boundaries and geometric patterns are insufficiently highlighted. This indicates that minimizing the discrepancy between $M_i$ and $M_{\text{multi}}$ alone cannot guarantee that the learned anchor map captures meaningful cross-modal structural information. After introducing $L_{prior}$, the anchor maps exhibit substantially stronger responses along stable scene structures, including building boundaries, road layouts, vegetation interfaces, and terrain discontinuities. The consistent improvement across the four datasets indicates that the structural prior suppresses modality-specific appearance responses and promotes shared geometric cues. Furthermore, incorporating $L_{tv}$ further suppresses isolated responses and irregular texture patterns, producing more spatially continuous and coherent anchor maps. Therefore, $L_{tv}$ serves as a complementary spatial

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TABLE III

QUANTITATIVE ABLATION STUDY OF DIFFERENT LOSS FUNCTIONS IN THE PROPOSED ANCHORMAP EXTRACTOR. **BOLD** AND UNDERLINED VALUES INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

| Loss | | | FUSAR-MaP | | | Hunan-DataSet | | | DFC2019 | | | MDAS | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $L_{distill}$ | $L_{prior}$ | $L_{tv}$ | Edge Corr↑ | MAE↓ | TV↓ | Edge Corr↑ | MAE↓ | TV↓ | Edge Corr↑ | MAE↓ | TV↓ | Edge Corr↑ | MAE↓ | TV↓ |
| ✓ | □ | □ | -0.0806 | **0.0003** | **0.0002** | -0.0138 | **0.0001** | **0.0001** | 0.0640 | **0.0008** | **0.0007** | -0.0236 | **0.0003** | **0.0000** |
| ✓ | ✓ | □ | 0.9487 | 0.0091 | 0.0119 | 0.9817 | 0.0092 | 0.0101 | 0.9611 | 0.0055 | 0.0079 | 0.9644 | 0.0092 | 0.0078 |
| ✓ | ✓ | ✓ | **0.9665** | 0.0045 | 0.0053 | **0.9845** | 0.0020 | 0.0025 | **0.9705** | 0.0034 | 0.0045 | **0.9823** | 0.0033 | 0.0032 |

TABLE IV

OVERALL QUANTITATIVE ABLATION STUDY OF THE PROPOSED FRAMEWORK ON FOUR BENCHMARK DATASETS. **BOLD** AND UNDERLINED VALUES INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

| | Structural Condition | | | Condition Composition | | | Guidance Scheme | | | FUSAR-MaP | | | Hunan-DataSet | | | DFC2019 | | | MDAS | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| No. | OPT | Canny | Anchor | Single | Masked Fusion | | Attention | Structure | Guide Scale | FID↓ | KID↓ | NMI↑ | FID↓ | KID↓ | NMI↑ | FID↓ | KID↓ | NMI↑ | FID↓ | KID↓ | NMI↑ |
| 1 | ✓ | □ | □ | ✓ | □ | | □ | ✓ | 0.5 | 141.335 | 0.0886 | 0.1342 | 247.916 | 0.0641 | 0.1532 | 213.900 | 0.0392 | 0.1639 | 82.910 | 0.0401 | 0.1552 |
| 2 | □ | ✓ | □ | ✓ | □ | | □ | ✓ | 0.5 | 140.297 | 0.0854 | 0.1223 | 246.885 | 0.0623 | 0.1497 | 212.436 | 0.0371 | 0.1465 | 83.002 | 0.0414 | 0.1453 |
| 3 | □ | □ | ✓ | ✓ | □ | | □ | ✓ | 0.5 | 141.302 | 0.0873 | 0.1266 | 245.274 | 0.0577 | 0.1521 | 213.268 | 0.0390 | 0.1571 | 82.948 | 0.0406 | 0.1480 |
| 4 | ✓ | □ | ✓ | □ | ✓ | | □ | ✓ | 0.5 | 140.609 | 0.0821 | 0.1381 | 234.460 | 0.0561 | 0.1742 | 201.721 | 0.0325 | 0.1676 | 79.925 | 0.0398 | 0.1594 |
| 5 | ✓ | □ | ✓ | □ | ✓ | | □ | ✓ | 0.3 | 86.127 | **0.0495** | 0.1347 | 228.475 | 0.0494 | 0.1510 | 193.080 | 0.0287 | 0.1407 | 62.213 | **0.0237** | 0.1568 |
| 6 | ✓ | □ | ✓ | □ | ✓ | | □ | ✓ | 1.0 | 183.428 | 0.1321 | 0.1415 | 262.396 | 0.0561 | **0.1957** | 217.490 | 0.0357 | 0.1705 | 103.572 | 0.0641 | **0.1642** |
| 7 | ✓ | □ | ✓ | □ | ✓ | | ✓ | ✓ | SA-Hybrid | **86.095** | 0.0508 | **0.1419** | **227.319** | **0.0463** | 0.1951 | **191.088** | **0.0261** | **0.1707** | **61.002** | 0.0243 | 0.1638 |

regularization term that improves the structural completeness of the extracted anchor representation.

The quantitative results in Table III further validate the effectiveness of each loss term. When only $L_{distill}$ is employed, the extracted anchor maps achieve low MAE and TV values but exhibit poor Edge Corr, indicating that minimizing the prediction discrepancy alone may result in over-smoothed responses without meaningful structural information. After introducing $L_{prior}$, the Edge Corr is significantly improved across all datasets, demonstrating that the structural prior effectively guides the extractor toward cross-modal structural patterns. Furthermore, incorporating $L_{tv}$ improves the spatial coherence while preserving structural responses, leading to the highest Edge Corr with low MAE and TV values. These results indicate that the complete objective achieves a better trade-off between structural validity, cross-modal consistency, and spatial smoothness.

*2) Ablation of AnchorDiff Framework Components:* After determining the optimal training configuration of the AnchorMap Extractor, we further evaluate the contribution of each proposed component in the complete AnchorDiff framework. Specifically, seven variants are designed to analyze three key designs: 1) the structural condition provided by the anchor map; 2) the Masked Fusion strategy; and 3) the SA-Hybrid scheme. All experiments are conducted on four datasets and evaluated with the same metrics to ensure fair and consistent comparisons. These variants are denoted as No. 1–No. 7 in Table IV.

Specifically, No. 1 uses OPT as the structural condition with single structural guidance, while No. 2 and No. 3 replace it with the Canny edge map and the proposed anchor map, respectively. No. 4 introduces Masked Fusion for condition composition. Based on the same Masked Fusion setting, No. 4–No. 6 perform single-stage sampling with different fixed structural guidance scales, whereas No. 7 adopts the complete two-stage SA-Hybrid strategy with a weak-guidance reference pass and a strong-guidance refinement pass. These comparisons allow us to separately analyze the effects of structural condition, condition composition, and sampling strategy on generation fidelity and cross-modal consistency.

*a) Effect of structural conditions:* We first compare No. 1–No. 3 to investigate the impact of different structural conditions on cross-modal generation. As reported in Table IV, using OPT as the structural condition in No. 1 provides certain structural information, but its performance on cross-modal consistency remains limited. This is mainly because OPT contains modality-dependent appearance variations and redundant semantic responses, which may introduce ambiguous guidance during generation.

The Canny edge map used in No. 2 enhances local edge responses and provides more explicit boundary constraints. However, due to the lack of high-level structural semantics and global scene representation, it fails to establish reliable structural correspondence across modalities, resulting in degraded generation quality and unstable spatial layouts.

In contrast, the proposed anchor map in No. 3 achieves consistent improvements in both cross-modal consistency and generation fidelity. Unlike raw OPT features or hand-crafted edge maps, the anchor map is specifically designed to extract modality-shared scene structures while suppressing modality-specific appearance interference. Therefore, it provides more reliable structural priors for target-modality generation and effectively improves structural alignment across modalities.

The qualitative comparisons in Fig. 6 further demonstrate that AnchorMap-based guidance better preserves global scene layouts and significantly reduces structural inconsistencies in the generated results. These observations validate the superiority of anchor maps as an effective structural condition for cross-modal generation tasks.

*b) Effect of masked fusion:* We further compare No. 1, No. 3, and No. 4 to evaluate the effectiveness of the proposed Masked Fusion strategy. Compared with single structural guidance, No. 4 further introduces the proposed Masked Fusion strategy, which leverages the anchor map as a mask over OPT to reinforce the guidance. The results show that No. 4 consistently improves over No. 1 and No. 3 across the evaluation metrics. This indicates that Masked Fusion is more

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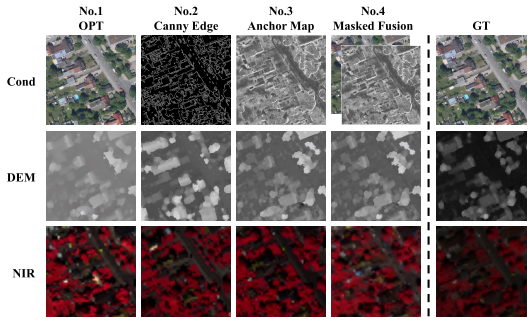



Fig. 6. Qualitative comparison of different structural conditions and the proposed AnchorMap-based masked fusion strategy across multiple modalities on MDAS.

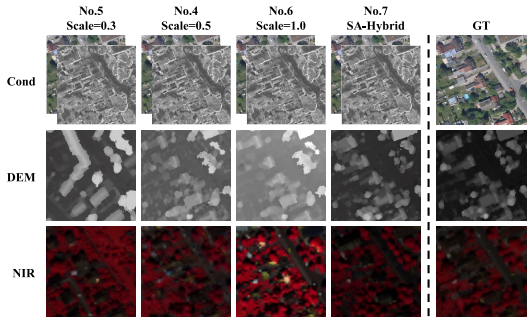



Fig. 7. Qualitative comparison between single-stage sampling with different fixed structural guidance strengths and the proposed two-stage SA-Hybrid across multiple modalities on MDAS.

effective than direct structural guidance for condition fusion. By introducing a spatially selective fusion mechanism, Masked Fusion helps the model better utilize useful structural cues during target-modality generation, thereby improving both generation quality and cross-modal consistency. As observed in Fig. 6, Masked Fusion also produces more stable spatial layouts and fewer visually noticeable structural deviations.

*c) Effect of guidance strength and SA-Hybrid:* We further compare No. 4–No. 7 to investigate the trade-off between structural guidance and modality-specific fidelity and to verify the proposed two-stage sampling strategy. No. 4–No. 6 use single-stage sampling with fixed guidance scales of 0.5, 0.3, and 1.0, respectively. As the guidance strength increases, NMI consistently improves across the four datasets, indicating stronger cross-modal structural alignment, whereas FID and KID generally deteriorate because excessive structural constraints suppress target-modality characteristics. This trade-off motivates the proposed two-stage design. In No. 7, the weak-guidance pass first establishes a fidelity-aware attention reference, while the strong-guidance pass strengthens structural constraints and is regularized by the cached reference attention. Consequently, SA-Hybrid retains structural consistency comparable to strong single-stage guidance while substantially recovering generation fidelity. The qualitative results in Fig. 7 further confirm this behavior, showing that the proposed strategy preserves coherent cross-modal structures without the appearance degradation observed under excessive single-stage guidance.

Overall, the ablation results validate the effectiveness of the main components in AnchorDiff. The anchor map provides a more reliable structural condition for cross-modal generation, Masked Fusion improves the effectiveness of condition fusion, and SA-Hybrid further balances structural consistency and generation quality.

TABLE V

INFERENCE TIME, GPU MEMORY USAGE AND PARAMETER COUNT COMPARISON OF DIFFERENT METHODS UNDER THE SAME EXPERIMENTAL SETTING

| Method | Inference Time (s/img) | GPU Usage (GB) | Params (M) |
|---|---|---|---|
| CycleGAN [24] | 5.43 | 3.56 | 11.38 |
| Pix2Pix-Zero [27] | 12.23 | 14.93 | 2132.47 |
| CRS-Diff [14] | 8.79 | 19.42 | 2493.75 |
| AnchorDiff | 16.92 | 19.42 | 2495.29 |

### D. Computational Efficiency

To further evaluate the practical applicability of the proposed method, we compare the computational efficiency, GPU memory consumption, and parameter size of AnchorDiff with representative methods. All methods are evaluated under the same experimental setting. The average inference time per generated image, peak GPU memory usage, and model parameter size are reported in Table V.

As shown in Table V, GAN-based methods achieve faster inference and lower memory consumption due to their feed-forward generation paradigm. CycleGAN [24], for example, requires only 5.43 s per image and 3.56 GB of GPU memory, with a compact parameter size of 11.38 MB. However, as demonstrated in the previous quantitative and qualitative comparisons, such methods are less effective in preserving fine-grained cross-modal spatial consistency. In contrast, diffusion-based methods generally require higher computational costs because of the iterative denoising process and their larger generative backbones, but they provide stronger generative capacity for complex remote sensing scenes.

Compared with CRS-Diff [14], AnchorDiff adopts the same diffusion backbone, resulting in the same peak GPU memory usage of 19.42 GB. The parameter size increases only marginally from 2493.75M to 2495.29M due to the lightweight shared-region extractor. This indicates that the proposed anchor-guided generation framework introduces only a negligible number of additional learnable parameters, approximately 1.54M, compared with the diffusion backbone. The increased inference time mainly comes from the progressive injection of shared structural constraints during the denoising process. Specifically, AnchorDiff requires 16.92 s per image, which is less than twice that of CRS-Diff [14]. Therefore, AnchorDiff achieves improved image fidelity and cross-modal spatial alignment with only moderate inference-time overhead, while maintaining the same peak GPU memory usage and adding only a negligible parameter overhead compared with the baseline diffusion model.

### E. Downstream Applicability

Beyond standard text-to-aligned multimodal generation, we further investigate the practical applicability of AnchorDiff in several downstream scenarios. In real-world remote sensing applications, multimodal datasets often differ in spatial

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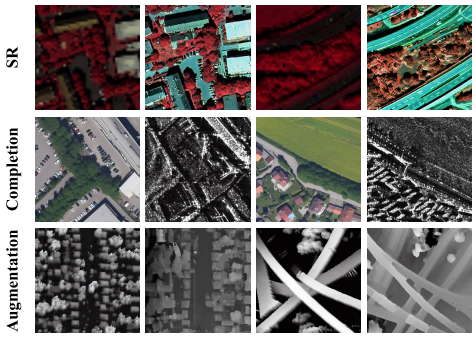



Fig. 8. Qualitative results of downstream applications, including cross-dataset super-resolution, missing-modality completion, and aligned multimodal data augmentation.

resolution, modality composition, and sample scale. A practically useful multimodal generator is therefore expected not only to synthesize high-quality aligned modalities within a single dataset, but also to support cross-dataset resolution enhancement, missing-modality completion, and dataset augmentation. To this end, we conduct three representative application studies, including NIR super-resolution, SAR modality completion, and aligned multimodal data augmentation. The qualitative results are shown in Fig. 8.

*1) Cross-Dataset Super-Resolution:* The MDAS and DFC2019 datasets both contain NIR imagery, but their NIR modalities have different spatial resolutions. This provides a natural scenario for evaluating whether the high-resolution distribution learned from one dataset can be transferred to another dataset. Specifically, we use the model trained on the higher-resolution DFC2019 data to enhance the NIR modality of MDAS. By extracting a modality-shared structural anchor from the available MDAS scene information, AnchorDiff can preserve the original scene layout while generating sharper high-frequency details in the target NIR image. As shown in the first row of Fig. 8, the generated NIR results exhibit clearer boundaries of roads, buildings, and vegetation regions, suggesting that AnchorDiff can serve as a cross-dataset resolution enhancement tool when high-resolution target-modality observations are limited.

*2) Missing-Modality Completion:* Another common limitation of multimodal remote sensing datasets is incomplete modality coverage. For example, FUSAR-MaP provides SAR observations, whereas MDAS does not contain the SAR modality. We therefore explore whether AnchorDiff can synthesize a plausible SAR modality for MDAS scenes by transferring the cross-modal structural relationship learned from SAR-containing datasets. Given the shared anchor derived from the available modalities, AnchorDiff generates SAR-like images that preserve the major spatial structures of the target scene while introducing modality-specific backscattering characteristics. As illustrated in the second row of Fig. 8, the synthesized SAR images maintain coherent road layouts, land-cover boundaries, and urban structures. This indicates that AnchorDiff can be used as a missing-modality completion strategy for constructing more complete multimodal remote sensing data.

*3) Cross-Dataset Same-Modality Data Augmentation:* We further evaluate the potential of AnchorDiff for cross-dataset same-modality data augmentation. Compared with MDAS, DFC2019 contains a relatively limited number of training samples, which may restrict the diversity of the learned elevation-related data distribution and lead to inferior generation quality when directly using DFC2019 for data augmentation. To examine whether AnchorDiff can benefit from a larger dataset under the DEM modality category, we compare the generated elevation maps obtained from models trained on DFC2019 and MDAS, respectively. It is worth noting that the DEM modality in DFC2019 corresponds to above-ground-level (AGL) height maps, whereas MDAS provides digital surface model (DSM) data. As shown in the third row of Fig. 8, the samples generated from the MDAS-trained model exhibit better visual continuity, smoother spatial transitions, and more coherent structural patterns than those generated from the DFC2019-trained model. These results indicate that AnchorDiff can leverage larger same-modality datasets to improve the quality of synthetic elevation data, providing a practical way to augment scarce DEM-related remote sensing datasets.

Overall, the downstream experiments demonstrate that AnchorDiff is not limited to the standard multimodal generation setting. By explicitly modeling modality-shared structural anchors, the proposed method can be flexibly extended to cross-dataset super-resolution, missing-modality completion, and cross-dataset same-modality data augmentation. This further validates its practical value for multimodal remote sensing scenarios where high-quality, spatially aligned, and modality-complete data are difficult to acquire.

## V. Conclusion

In this article, we proposed AnchorDiff, a novel framework for text-to-aligned multimodal remote sensing image generation. The core innovation of AnchorDiff is the introduction of a modality-shared structural anchor, which serves as a common spatial reference to guide the generation of multiple modalities while preserving their unique characteristics. The proposed AnchorMap Extractor effectively captures the shared structural information across heterogeneous remote sensing modalities, while the SA-Hybrid scheme further enhances the generation quality and cross-modal consistency. Extensive experiments on four benchmark datasets demonstrate that AnchorDiff significantly outperforms state-of-the-art methods in terms of image fidelity, semantic consistency, and spatial alignment. A remaining limitation is that structural errors in the generated source image may propagate through the predicted anchor map and affect subsequent target-modality generation. Future work will investigate multisource or confidence-aware anchor estimation to improve robustness against such source-generation errors.

The proposed method also shows promising applicability in downstream tasks such as cross-dataset super-resolution, missing-modality completion, and aligned multimodal data augmentation. Future work will explore controllable multimodal remote sensing generation, more efficient latent-space

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sampling strategies, and broader applications in scalable multimodal remote sensing analysis and interpretation.

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**Yuhan Kang** received the bachelor's degree from the College of Science, Sichuan Agricultural University, Yaan, China, in 2020. He is currently pursuing the Ph.D. degree with Hunan University, Changsha, China.

His research interests include computer vision and machine/deep learning applications, especially image generation, object detection, and domain adaptation.



**Honghu Pan** received the Ph.D. degree from the School of Computer Science and Technology, Harbin Institute of Technology, Shenzhen, Shenzhen, China, in 2023.

From January 2024 to 2026, he was a Post-Doctoral Researcher with the University of Macau, Macau, China. He is currently an Assistant Professor with the School of Artificial Intelligence and Robotics, Hunan University, Changsha, China. His research interests include computer vision, generative models, and multimodal large models.



**Hao Liu** (Graduate Student Member, IEEE) received the bachelor's and master's degrees in electronic information engineering from Xidian University, Xi'an, China, in 2021 and 2024, respectively. He is currently pursuing the Ph.D. degree with the Department of Information Engineering and Computer Science, University of Trento, Trento, Italy.

His research interests include applying deep learning algorithms to facilitate the fusion and interpretation of multisource remote sensing data.



**Qiang Liu** received the B.S. degree from Taiyuan University of Technology, Taiyuan, China, in 2017, and the M.S. degree from Shenzhen University, Shenzhen, China, in 2020. He is currently pursuing the Ph.D. degree in control science and engineering with Hunan University, Changsha, China.

His research interests include computer vision and weakly-supervised learning.



**Weiying Xie** (Senior Member, IEEE) received the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in 2017.

She is currently a Professor with the State Key Laboratory of Integrated Services Networks, Xidian University. She has authored more than 50 articles in refereed journals and proceedings, including IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON CYBERNETICS, and Conference on CVPR and AAAI conferences. Her research interests include neural networks, machine learning, hyperspectral image processing, and high-performance computing.



**Leyuan Fang** (Senior Member, IEEE) received the Ph.D. degree from the College of Electrical and Information Engineering, Hunan University, Changsha, China, in 2015.

From September 2011 to 2012, he was a Visiting Ph.D. Student with the Department of Ophthalmology, Duke University, Durham, NC, USA, supported by China Scholarship Council. From August 2016 to September 2017, he was a Post-Doctoral Researcher with the Department of Biomedical Engineering, Duke University. He is a Professor with the School of Artificial Intelligence and Robotics, Hunan University, and an Adjunct Researcher with the Pengcheng Laboratory, Shenzhen, China. His research interests include sparse representation and multiresolution analysis in remote sensing and medical image processing.

Dr. Fang is an Associate Editor for IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, and IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS.



**Lorenzo Bruzzone** (Fellow, IEEE) received the Laurea (M.S.) degree (summa cum laude) in electronic engineering and the Ph.D. degree in telecommunications from the University of Genoa, Genoa, Italy, in 1993 and 1998, respectively.

He is currently a Full Professor of telecommunications with the University of Trento, Trento, Italy, where he teaches remote sensing, radar, and digital communications. He is the Founder and the Director of the Remote Sensing Laboratory, Department of Information Engineering and Computer Science, University of Trento. He is a principal investigator of many research projects. Among others, he is also a Principal Investigator of the Radar for Icy Moon Exploration (RIME) Instrument in the framework of the Jupiter Icy Moons Explorer (JUICE) Mission of the European Space Agency (ESA) and the Science Lead of the High Resolution Land Cover Project in the framework of the Climate Change Initiative of ESA. He is the author (or co-author) of 294 scientific publications in refereed international journals (221 in IEEE journals), more than 340 papers in conference proceedings, and 22 book chapters. He is the editor/co-editor of 18 books/conference proceedings and one scientific book. His papers are highly cited, as proved by the total number of citations (more than 41 000) and the value of the H-index (95) (source: Google Scholar). He was invited as a keynote speaker at more than 40 international conferences and workshops. His research interests include remote sensing, radar and synthetic aperture radar (SAR), signal processing, machine learning, and pattern recognition. He promotes and supervises research on these topics within the frameworks of many national and international projects.

Dr. Bruzzone has been a member of the Administrative Committee of the IEEE Geoscience and Remote Sensing Society (GRSS) since 2009, where he has been the Vice-President of professional activities since 2019. He ranked First Place in the Student Prize Paper Competition of the 1998 IEEE International Geoscience and Remote Sensing Symposium (IGARSS), Seattle, July 1998. He was a recipient of many international and national honors and awards, including the recent IEEE GRSS 2015 Outstanding Service Award, the 2017 and 2018 IEEE IGARSS Symposium Prize Paper Awards, and the 2019 WHISPER Outstanding Paper Award. He was a Guest Coeditor of many special issues of international journals. He is the Co-Founder of the IEEE International Workshop on the Analysis of Multi-Temporal Remote-Sensing Images (MultiTemp) series and is currently a member of the Permanent Steering Committee of this series of workshops. Since 2003, he has been the Chair of the SPIE Conference on Image and Signal Processing for Remote Sensing. He has been the Founder of *IEEE Geoscience and Remote Sensing Magazine*, for which he was the Editor-in-Chief from 2013 to 2017. He is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. He was a Distinguished Speaker of the IEEE Geoscience and Remote Sensing Society from 2012 to 2016.

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